11 Jack and Sadia work for a company that sells boxes of breakfast cereal.

The company wants to have a special offer.

Here is Jack's idea for the special offer.

> Put 25% more cereal into each box and do **not** change the price.

Here is Sadia's idea.

> Reduce the price and do **not** change the amount of cereal in each box.

Sadia wants her idea to give the same value for money as Jack's idea.

By what percentage does she need to reduce the price?

Choose an ordinary box with 100 g costing 100 p.

Jack offers 125 g for the same 100 p.

1 mark
Represent the 25% extra cereal

	Grams	Pence	
	125	100	
÷5	25	20	÷5
×4	100	80	×4

1 mark
Find the equivalent price of 100 g

Equivalent price = 80 p for 100 g.

Reduction = 100 p – 80 p = 20 p

20 p out of 100 p is a 20% reduction.

1 mark
Same cereal-to-price ratio

20 ..... %

(Total for Question 11 is 3 marks)